Homework 1

This Homework is due on Wednesday, September 17 by 11:59pm. Please show all your work and write as clear as possible.

1. A digital scale that provides weights to the nearest gram is used.

   (a) What is the sample space for this experiment?

   Let A denote the event that weight exceeds 11 grams, let B denote the event that weight is less than or equal to 15 grams, and let C denote the event that a weight is greater than or equal to 8 grams and less than 12 grams. Describe the following events:

   (b) $A \cup B$

   (c) $A \cap B$

   (d) A'

   (e) $A \cup B \cup C$

   (f) $(A \cup B)'$

   (g) $A \cap B \cap C$

   (h) $B' \cap C$ i

   (i) $A \cup (B \cap C)$

2. A soccer player on the GMU soccer team has a probability of scoring on a penalty kick against a particular goalie of $p = 0.80$. Suppose in the course of a game, she has 3 penalty kicks against the goalie.

   (a) What is the probability she scores on all 3 kicks (assuming independence)?

   (b) What is the probability she fails to score on all 3 kicks?

3. A computer has 4 independent backup drives. Each drive fails with probability 0.02. What is the probability that the computer does not lose all its backups?

4. Fill in the blanks of the contingency table below. The contingency table reflects the result of a survey of **200** individuals on their preferred category of beer.

	Light	Regular	Dark	Total
Male			50	
Female	20			95
Total	50		80	

   (a) What's the probability of being a female and liking dark beer?

   (b) Conditional on being a male, what's the probability of liking light beer?

5. A study of the effect of parents' smoking habits on the smoking habits of students in a high school produced the following table of proportions. Suppose a student is randomly selected from this population.

Parents	Student smokes	Student does not smoke	Total
Both parents smoke	0.07	0.26	0.33
One smokes	0.08	0.34	0.42
Neither smokes	0.03	0.22	0.25
Total	0.18	0.82	1.00

   (a) What is the probability the student smokes and at least one parent of the student smokes?

   (b) Given the student smokes, what is the probability neither parent smokes?

6. You are on a trip to Vegas and decide to play the slot machines. The casino has installed "good" and "bad" slot machines where the probability of winning on a good machine is 75% while the probability of winning on a bad machine is only 15%. After reading an exposé on the casino industry in the inflight magazine you know that 80% of the slot machines are "bad" at this particular casino. If you play once and you win, what is the probability that you are playing on a good machine?

7. Chicago food establishments are classified by 3 levels of Risk (High, Medium, and Low). The proportions (probabilities) are: P(High) = .65, P(Medium) = .22, P(Low) = .13. The probabilities of Failing inspection are .21 among High risk, .23 among Medium Risk, and .33 among Low Risk.

   (a) Compute the probability a randomly selected establishment Fails inspection.

   (b) Compute the probability of each risk type, given the establishment fails inspection.